

**Kathmandu University School of Education
Department of STEAM Education**

**Bachelor of Technical Education (B Tech Ed) Program
Information Technology (IT)**

In Association with School of Engineering, KU

Course Curriculum

Semester V

2022

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Overview of the B Tech Ed Program

Technical and Vocational Education and Training (TVET) has been one of the prominent sectors globally for a long. Dewey (1916) saw TVET as a tool for education reform in modernizing society. This has been equally vibrant in the present context as scholars (e.g., Maclean & Wilson, 2009) argue that TVET is an education and training that prepares an individual for gainful employment. TVET programs equip young people with the skills, knowledge, and competencies required to enter a particular type of professional career (OECD 2017). Hence, the role of TVET in preparing skilled human resources and enabling them to transition into the career path for gainful employment in specific and the country's economic development in a broad is inevitable. The formal TVET began in Nepal only around mid of the 20th century. Nevertheless, occupational skills have been transformed from generation to generation for a long (Ministry of Education [MoE], 2012). The establishment of the Council for Technical Education and Vocational Training (CTEVT), the governing body of TVET in 1989, was a substantive effort toward its development and expansion (CTEVT, 2019). Additionally, at present, the Centre for Education and Human Resource Development (CEHRD) offering technical diplomas in 284 technical stream schools throughout the country (GoN, 2018) and universities (such as Kathmandu University, Manmohan Technical University, etc.) run bachelor and master in TVET and other 11 federal ministries offer TVET related formal and non-formal programs (MoEST, 2019). The CTEVT, which is in charge of formulating TVET-relevant policies, developing standards for programs and curriculums, coordination, accreditation, monitoring, and supervision (CTEVT, 2019) largely shares the TVET related activities. There are 1131 CTEVT affiliated and constituent institutions with roughly the average capacity of 70,000 per annual operate formal TVET programs (CTEVT, 2020). Further expansion of TVET institutions in the country is yet to be expected soon since the government has a provision of establishing at least one technical school in all local levels.

Bachelor of Technical Education is a program to develop instructors and teachers in the field of Technical and Vocational Education and Training (TVET) with a view to fulfil the gap of acute shortage of instructors and TEVT professionals in the country which aspires to promote technical and vocational education two-third of the educational system and processes. The program has envisaged that a mix of two or more of the work-based learning, activity-based instruction, on the job training, project-based learning and problem-based learning be used to deliver the program. Because of the 'practical focus' of the program, the assessment has been proposed to address the needs of experiential learning.

The 136-credit program has been structured four types of the courses, such as Educational Core (18), Vocational Specialization (80), Vocational Pedagogy (20), and Educational Research and Internship (18) with a view to inculcate both pedagogical and vocational skills. The vocational specializations have been offered in a manner that they can be utilize hands-on activities as well. The program begins with the foundational courses in general education, vocational pedagogy and the area of specialization. In the later part of the program, OJT/WBL and forms of experiential learning shall be introduced. The culmination of the program will be internships in their specializations and a research report that addresses the problem of vocational education.

B Tech Ed in Information Technology

Bachelor of Technical Education is a 4-year (8 semesters) program to develop instructors, teachers, professionals in the field of Technical and Vocational Education and Training (TVET) with a view to fulfil the gap of acute shortage of instructors and TEVT professionals in the country which aspires to promote technical and vocational education two-third of the educational system and processes. The program has envisaged that a mix of two or more of the work-based learning, activity-based instruction, on the job training, project-based learning and problem-based learning be used to deliver the program. Because of the ‘practical focus’ of the program, the assessment has been proposed to address the needs of experiential learning. The B Tech Ed in IT program aims at providing technical, vocational, and educational exposures for students who can become IT professionals, IT Instructors, TVET educators, TVET curriculum developers, etc. The occupational/specialization courses emphasize the development of core IT skills in the field of computer science and technology innovation. The B Tech Ed in IT graduates will be eligible for doing master's program in IT, TVET, Education, Social Sciences, etc.

Major Objectives of the B Tech Ed in IT Program

- Demonstrate Level 6 competencies envisaged by the National Vocational Qualification Framework (NVQF) of Nepal,
- Exhibit comprehensive, meaningful, and coherent knowledge and skills in the field of technology and computer science
- Apply occupational specialization knowledge and skills in solving problems occurred in their contexts of work,
- Apply vocational pedagogical knowledge and skills in technical and vocational instruction of Information Technology,
- Show scholarly literacy, communication, quantitative reasoning, critical thinking, learning skills needed for advanced and work-based learning,
- Exemplify a deep and principled understanding of the technical and vocational learning processes and the role of the instructors in facilitating these processes in the students,
- Apply a wide range of teaching process skills including curriculum development, lesson planning, materials development, assessment, and pedagogical approaches, and
- Enable students to become resilient and lifelong learners by harnessing internal and external assets as TVET professionals.

Career Prospects

- TVET Instructors and Educators
- TVET Curriculum Developers
- IT Professionals (Software and Apps developer, programmer, Network Engineer, Web Developer, IT Admin, IT Engineer, etc.)
- Eligible for doing master's program in IT, TVET, Education, Social Sciences, etc.

Entry Requirements

The minimum prerequisite for this program is the completion of Level 5 education (as per National Vocational Qualification Framework) in relevant vocational specializations. The details are-

- Minimum 1.6 CGPA in 10+2
- Or
- A minimum 40% marks in 10+2, PCL or equivalent in any disciplines

Courses Offered in the Semester V

Course	Credit Hour		
	Theory	Practical	Total
EDUC 300 Curriculum Development	2	1	3
VPED 300 Designing occupational Curriculum	1	1	2
VPED 301 TVET Ecosystem in Nepal	1	1	2
ITEX 300 Management Information System	1	1	2
ITEX 320 Web Engineering	2	1	3
ITEX 321 Software Engineering	1	1	2
ITEX 301 System Network Administration	1	1	2
ITEX 340 Professional IT project I	1	1	2
Total	10	8	18

Core Courses

EDUC 300 Curriculum Development

Course Name: Curriculum Development

Code: EDUC 300

Credit Hour: 3 (2+1)

Nature: Theory + Practical

Course Description and Goals

EDUC 300 - Curriculum Development is a foundational course designed for undergraduate students. This course provides a comprehensive introduction to the principles, theories, and practical aspects of curriculum development, tailored to meet the needs of both educational theorists and practical implementers. Throughout this course, students will explore the concept of 'curriculum' from multiple perspectives, examining its role in shaping educational experiences and outcomes. They will delve into historical and contemporary theories of curriculum design to understand how educational goals are translated into structured learning experiences. Key models of curriculum development, such as Tyler's Rationale and Taba's Model, will be studied to highlight different approaches to curriculum planning and execution. The course also emphasizes the importance of aligning curriculum with educational standards and learner needs. Students will learn how to develop sample curriculum based on the guidelines and curriculum development models. A significant component of the course involves practical application. Students will engage in a hands-on project where they design and evaluate a curriculum. This project will allow students to apply theoretical knowledge to real-world educational challenges, fostering skills in critical thinking, creativity, and problem-solving. By the end of the course, students will have gained the skills necessary to design, implement, evaluate, and refine effective curricula across a variety of educational contexts.

Learning Outcomes

The following are the learning outcomes that students will develop:

- Define key concepts and principles of curriculum development.
- Analyze curriculum development process with its major focus on practical aspects of curriculum development.
- Design/Develop the sample curriculum based on some major curriculum design framework and models.
- Evaluate the major ideas in curriculum implementation practice, including the support and issues in curriculum implementation.
- Evaluate and analyze the curriculum change process, including the major factors affecting curriculum development and change.

Content with specific objectives

Specific objectives	Contents
• To define curriculum with its etymological and other meanings • To differentiate among curriculum, syllabus, and textbook • To describe, illustrate, and exemplify the major sources of	Module I: Introduction to Curriculum (6 Hours) • Meaning and Concept of curriculum • Different Concepts of Curriculum • Hidden Curriculum • Curriculum, Syllabus, and Textbook • Sources of Curriculum

curriculum with its impact on curriculum development process	○ Historical and Philosophical Foundations ○ The Learners ○ Major Learning theories (Behaviorism, Cognitivism, & Constructivism) ○ The nature of knowledge ○ Society and Environment
• To illustrate and describe the major component of curriculum development process • To create samples of each curriculum development element and discuss with examples • To critically evaluate each step and element of curriculum development process and try to generate other possible strategies for curriculum development	Module II: Curriculum Development Process (10 Hours) • The components of Curriculum development ○ Need Assessment and Diagnosis ○ Selection of Learning Objectives (Meaning; concept of Aims, Goals, and Objectives; Bloom's taxonomy for objective development) ○ Learning Contents (Meaning of content, Relation between content and curriculum objectives, selection and organization of contents) ○ Learning Experience (Meaning, Contents Vs. Learning Experience, Selection and organization of learning experience) ○ Evaluation of Curriculum Outcomes
• To describe and illustrate the major characteristics and features of discipline-centered, learner-centered, and problem-centered design of curriculum with their major implications in education • To define and interpret the different curriculum design and development models (Tyler's Rationale Model, Hilda Taba's Model, and Integrated Curriculum) • To define and exemplify Integrated Curriculum (Multi-disciplinary, Interdisciplinary, and Transdisciplinary) • To design the curriculum framework with major components of the curriculum	Module III: Curriculum Design Models (18 Hours) • Discipline Centered Design • Learner Centered Design • Problem Centered Design • Ralph Tyler's Rationale Model of Curriculum Development • Hilda Taba's Curriculum Development Models • Integrated Curriculum Design ○ Multi-disciplinary ○ Interdisciplinary ○ Transdisciplinary
• To define and exemplify the curriculum implementation with its meaning • To describe various support required in curriculum implementation (such as Training	Module IV: Curriculum Implementation (6 Hours) • Meaning of Curriculum implementation • Support for curriculum implementation (Training and Faculty development, financial support, Peer-support, Monitoring support, Resource

and Faculty development, financial support, Peer-support, Monitoring support, Resource material development support, administrative support) • To define and discuss the important of communication skills in curriculum implementation • To describe and illustrate various issues and problems in the process of curriculum implementation	material development support, administrative support) • Communication in Curriculum implementation • Issues in Curriculum Implementation ○ Resistance to change of curriculum ○ Inadequate resources and support in professional development ○ Inadequacy of monitoring and supervision • ○ Lack of administrative and communication support ○ Inadequate plan and weak implementation strategies ○ Irrelevant Curriculum
• To analyze and evaluate the curriculum • To evaluate and describe various factors affecting the curriculum changes	Module IV: Curriculum Evaluation and Change (5 Hours) • Meaning and Need for Curriculum change • Evaluation of Curriculum • Factors affective curriculum changes ○ Technological Advancement ○ Economic Factors ○ Social Changes ○ Policy and Legislations ○ Cultural Factors ○ Globalization ○ Educational Research and Publications ○ Political Influence ○ Feedback from Educational Communities

Major Teaching and Learning Strategy

The following are the major teaching and learning strategies for facilitating the course contents effectively:

- Interactive Lectures
- Collaborative work and Discussions
- Questions-Answers sessions with students
- Presentations and Demonstration
- Discovery approach

Practical

The following practical work is recommended to perform:

Develop a framework or a sample curriculum based on one of the following curriculum design models:

- Discipline Centered Design
- Learner Centered Design
- Problem Centered Design
- Ralph Tyler's Rationale Model of Curriculum Development
- Hilda Taba's Curriculum Development Models
- Integrated Curriculum Design
 - Multi-disciplinary
 - Interdisciplinary
 - Transdisciplinary

Assessment Plan

- **In-semester (50)**
Assignment, written exam, presentation, case study, report, objectives, attendance
- **End-semester (50)**

Question Types	Marks Distribution	Total Question
MCQ	10 Marks	10 Questions
SAQ	24 Marks	7 Questions [Attempt 6 Questions]
LAQ	16 Marks	3 Questions [Attempt 2 Questions]

Marking Scheme for Final Examination:

- Section A (**MCQ**): 10 Questions $\times$ 1 marks
- Section B (**Short Answer Question-SAQ**) – Total 7 Questions $\times$ 4 marks
(Attempt any Six)
- Section C (**Long Answer Question-LAQ**) – Total 3 Questions $\times$ 8 marks
(Attempt any Two)

Learning Units	Total Credit Hours	MCQs	SQs	LQs	Remarks
Module I: Introduction to Curriculum	6	1	1		
Module II: Curriculum Development Process	10	2	1		1
Module III: Curriculum Design Models	18	4	3		1
Module IV: Curriculum Implementation	6	2	1		
Module IV: Curriculum Change	5	1	1		1

References:

- Aryal, P. N., & Niure, D. P. (2010). *Curriculum planning and practice*. Intellectuals Books Palace.
- Bhalla, N. (2007). *Curriculum development*. Authors Press.
- Marzano, R. J., & Kendall, J. S. (2007). *New taxonomy of educational objectives* (2nd Ed.). California: Carwin press.
- Ornstein, A. C., & Hunkins, F. P. (2004). *Curriculum foundations, principles and issues* (4th ed.). Pearson education. Inc.
- Murray, P. (1993). *Curriculum development and design*. Australia Allen and Unwin Pvt Ltd.
- Taba, H. (1962). *Curriculum development; theory and practice*. Brace and world, Inc.
- [Wiles, J., & Bondi, J. \(1993\). *Curriculum development; A guide in practice* \(4th Ed.\). Prentice Hall.](#)

Vocational Pedagogical Course

VPED 300 Designing Occupational Curriculum

Course Name: Designing Occupational Curriculum

Code: VPED 300

Credit Hour: 2 (1 TH + 1 PR)

Nature: Theory + Practical

Course Description and goals

This comprehensive course provides a detailed and extensive overview of the fundamental principles and best practices involved in designing occupational curricula. The primary goal of the course is to equip students with a deep understanding and proficiency in designing and developing occupational curricula. It covers a wide array of topics, including the theoretical foundations of curriculum design, conducting thorough needs analysis, exploring various curriculum design models, implementing effective instructional strategies, and utilizing diverse assessment techniques to ensure the efficacy of the curriculum.

Learning Outcomes

- Explore the definition and significance of curriculum design, tracing its historical evolution from ancient to modern perspectives.
- Evaluate various theories and models of curriculum design.
- Analyze the components of the occupational curriculum, including key elements such as goals, objectives, and standards.
- Conduct needs assessments effectively by identifying stakeholders and employing appropriate methods.
- Develop occupational curriculum content considering NOSS.
- Implement diverse assessment tools such as rubrics within occupational curriculum design, ensuring alignment with curriculum objectives and standards.
- Apply evaluation models like CIPP to assess the effectiveness of the occupational curriculum.

Content with specific objectives

Specific objectives	Contents
• To define curriculum design and highlight its importance. • To explore the historical perspectives of curriculum design. • To identify and analyze the key components of occupational curriculum.	Module I: Introduction to Curriculum Design (4 hours) 1.1. Definition and Importance of Curriculum Design 1.2. Historical Perspectives on Curriculum Design 1.2.1. <i>Ancient and Medieval Curriculum Design</i> 1.2.2. <i>Modern Curriculum Design</i> 1.3. Key Components of Occupation Curriculum

• To understand the principles and key concepts of Behavioral, Cognitive, and Constructivist theories in curriculum design. • To explore the components and processes of different curriculum models in curriculum design. • To apply Behavioral, Cognitive, and Constructivist theories and various curriculum models to occupational curriculum design.	Module II: Theories and Models of Curriculum Design (8 hours) 2.1. Behavioral, Cognitive, and Constructivist Theories 2.2. Curriculum Models 2.2.1. <i>Tylers' Model</i> 2.2.2. <i>Taba's Model</i> 2.2.3. <i>Backward Design Model</i> 2.2.4. <i>ADDIE Model</i> 2.3. Application of Theories and Models to Occupational Curriculum Design
• To define the concept of need assessment and distinguish between different types of need assessment. • To understand the importance of stakeholder analysis in curriculum design. • To explore various methods and techniques commonly used for need assessment. • To explore strategies for effective data collection and analysis.	Module III: Need Assessment and Analysis (6 hours) 3.1. Concept and Types of Need Assessment 3.2. Stakeholder Analysis 3.3. Methods and Techniques for Need Assessment 3.4. Data Collection and Analysis
• To understand the concept of occupational curriculum and identify the components of NOSS. • To explore the goals and objectives of occupational curriculum through the lens of Bloom's Taxonomy, SMART, and Occupational Standards and Competencies. • To identify methods for structuring occupational curriculum content effectively. • To explore the effectiveness of learning activities and instructional strategies.	Module IV: Occupational Curriculum and Its Development Process (8 hours) 4.1. Overview of Occupational Curriculum 4.1.1. <i>NOSS (National Occupational Skill Standard)</i> 4.2. Occupational Curriculum Goals and Objectives 4.2.1. <i>Bloom's Taxonomy</i> 4.2.2. <i>SMART objectives</i> 4.2.3. <i>Occupational Standards and Competencies</i> 4.3. Occupational Curriculum Content 4.4. Learning Activities and Instructional Strategies

• To understand the various types of assessment utilized in occupational curriculum development. • To explore different assessment tools commonly employed within the occupational curriculum. • To examine the application of rubrics in assessing occupational curriculum. • To understand the occupational curriculum evaluation model and familiarize students with the components of CIPP.	Module V: Assessment and Evaluation in Occupational Curriculum (4 hours) 5.1. Types of Assessments in Occupational Curriculum 5.2. Assessment Tools in Occupational Curriculum 5.2.1. <i>Rubrics</i> 5.3. Occupational Curriculum Evaluation Model 5.3.1 <i>CIPP (Context, Input, Process, Product)</i>
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Major Teaching and Learning Strategy (List of some Teaching and Learning Strategies for the faculty)

The following are the major teaching and learning strategies for facilitating the course contents effectively:

- Interactive and Guest Lectures
- Collaborative work and Discussions
- Questions-Answer sessions with students
- Presentations and Demonstration
- Discovery approach

Practical (List some of the practical and field-based tasks to be done)

1. Group Work: Students will work in groups to develop an occupational curriculum tailored to a specific industry or profession of their choosing. This multifaceted project will involve conducting thorough needs assessments to identify the specific requirements of the chosen industry, crafting precise learning outcomes, designing effective assessment methods, and creating engaging and informative instructional materials.

2. Individual Work: Students will be expected to visit a local vocational training center or Provincial CTEVT's curriculum division in a group. Students can observe and gain insights into the curriculum design (**DACUM**) process during this visit. Additionally, students can interact with instructors and stakeholders to further enrich their understanding of curriculum development in a real-world educational setting. After the visit, students must prepare a detailed field report individually documenting their observations, insights, and any recommendations for improving the occupational curriculum.

Assessment Plan

- **In-semester (50)**
Assignment, written exam, presentation, case study, report, objectives, attendance
- **End-semester (50)**

Question Types	Marks Distribution	Total Question
MCQ	10 Marks	10 Questions
SAQ	24 Marks	7 Questions [Attempt 6 Questions]
LAQ	16 Marks	3 Questions [Attempt 2 Questions]

Marking Scheme for Final Examination:

- Section A (**MCQ**): 10 Questions × 1 marks
- Section B (**Short Answer Question-SAQ**) – Total 7 Questions × 4 marks (Attempt any Six)
- Section C (**Long Answer Question-LAQ**) – Total 3 Questions × 8 marks (Attempt any Two)

Learning Units	Total Credit Hours	MCQs	SQs	LQs	Remarks
Module I: Introduction to Curriculum Design	4	1	1	0	
Module II: Theories and Models of Curriculum Design	8	3	2	1	
Module III: Need Assessment and Analysis	6	2	1	0.5	
Module IV: Occupational Curriculum and Its Development Process	8	3	2	1	
Module V: Assessment and Evaluation in Occupational Curriculum	4	1	1	0.5	

References:

- Gagné, R. M., Wager, W. W., Golas, K., & Keller, J. M. (2005). Principles of Instructional Design. Wadsworth.
- Ornstein, A. C., & Hunkins, F. P. (2018). Curriculum: Foundations, Principles, and Issues.
- Popham, W. J. (2017). Classroom Assessment: What Teachers Need to Know. Pearson.
- Stufflebeam, D. L., & Shinkfield, A. J. (2007). Evaluation Theory, Models, and Applications.
- Wiggins, G., & McTighe, J. (2005). Understanding by Design. ASCD.

VPED 301 TVET Ecosystem in Nepal

Course Name: TVET Ecosystem in Nepal

Code: VPED 301

Credit Hour: 2 (1 TH + 1 PR)

Nature: Theory + Practical

Course Description and goals

This course is designed to provide students with a comprehensive understanding of Nepal's Technical and Vocational Education and Training (TVET) ecosystems. It covers the structure, policies, stakeholders, and challenges within the TVET framework. The aim of this course is to familiarize students with different components of the TVET ecosystem in Nepal and around the world. Students will compare different TVET ecosystems across time and space and research TVET and industry connections by conducting case studies of the components of the TVET ecosystem. It will provide students with a comprehensive understanding of human resource management (HRM) principles and practices specific to the context of Technical and Vocational Education and Training (TVET) institutions. Moreover, this course aims to equip students with the necessary skills, knowledge, and attitudes to thrive in the workplace of tomorrow. Through theoretical learning and practical applications, students will gain the necessary knowledge and skills to contribute effectively to Nepal's TVET sector in the future.

Learning Outcomes

- Understand the concept and importance of TVET
- Develop a sound understanding of the different components of the TVET ecosystem
- Compare and contrast various TVET ecosystem models around the globe
- Understand the concepts and importance of National Vocational Qualification Framework (NVQF)
- Recognize the importance of industry collaboration in TVET, including the Dual VET Apprenticeship Model
- Understand the accreditation and certification system in TVET
- Understand and analyze the importance of HRM in TVET
- Explore the concept of the Future of Work and analyze the emerging trends in TVET

Content with specific objectives

Specific objectives	Contents
• To provide an overview of TVET, including its concepts and importance • To explore the historical background and evolution of TVET • To examine the evolution of TVET policies and regulations • To analyze the current status and trends in TVET	Module I: Introduction to TVET Education (4 hours) 1.1. Overview of TVET: Concepts and Importance 1.2. Historical background and evolution of TVET 1.3. Evolution of TVET policies and regulations 1.4. Current Status and Trends in TVET

• To familiarize students with the general landscape of TVET systems around the globe • To gain insights into different TVET systems across various countries • To establish a comprehensive understanding of TVET components • To identify and understand the roles and contributions of various stakeholders involved in TVET	Module II: TVET Ecosystems Worldwide (6 hours) 2.1. Overview of global TVET systems and structures 2.2. Comparison of TVET systems across different countries 2.3. Key components of TVET Ecosystems 2.3.1. <i>Government and Policy</i> 2.3.2. <i>Industry Engagement</i> 2.3.3. <i>TVET Institutions</i> 2.3.4. <i>TVET Instructors</i> 2.3.5. <i>Curriculum Development and Quality Assurance</i> 2.3.6. <i>TVET Financing</i> 2.4. Stakeholders in TVET Ecosystems 2.4.1. <i>Government agencies and ministries involved in TVET</i> 2.4.2. <i>Business Industry Associations (BIAs) and employers</i> 2.4.3. <i>Non-Governmental Organizations (NGOs) and International Agencies</i>
• To understand the structure and components of the education system in Nepal • To familiarize with the laws and regulations that govern TVET in Nepal • To gain insights into the various types of TVET institutions • To familiarize the concept of NVQF and understand the principles of lifelong learning, permeability, and equivalence in TVET • To explore the significance of Education-Employment Linkages (EEL) • To understand the accreditation standards, processes, and certification system	Module III: Framework of TVET in Nepal (10 hours) 3.1. Overview of Education System in Nepal 3.2. Legal framework governing TVET in Nepal 3.3. Overview of TVET institutions in Nepal 3.4. Concepts and Importance of National Vocational Qualification Framework (NVQF) 3.4.1. <i>Lifelong Learning in TVET</i> 3.4.2. <i>Permeability and Equivalence in TVET</i> 3.5. Linkages between TVET institutions and industries 3.5.1. <i>Importance of Industry Collaboration in TVET</i> 3.5.2. <i>Dual VET Apprenticeship Model in Nepal</i> 3.6. Accreditation and Certification System in TVET

	3.6.1. <i>Standards and Criteria for quality assurance in TVET</i> 3.6.2. <i>Accreditation processes for TVET institutions and programs</i>
• To emphasize the importance the HRM within TVET • To explore the processes and best practices involved in recruitment and selection for TVET institutions • To examine the role of training and development • To understand the significance of coaching, mentoring, and counseling • To explore the performance management practices • To discuss HR planning and strategic management within the framework of TVET	Module IV: TVET Human Resource Management (6 hours) 4.1. Definition and importance of HRM in TVET 4.2. Recruitment and Selection in TVET 4.2.1. <i>Job Analysis and Design in TVET</i> 4.2.2. <i>Recruitment methods and strategies for TVET institutions</i> 4.2.3. <i>Selection techniques and tools</i> 4.3. Training and Development in TVET 4.4. Coaching, mentoring, and counseling 4.5. Performance Management in TVET 4.6. HR Planning and Strategic Management in TVET
• To understand the concept of the Future of Work • To introduce students to soft skills with the evolving nature of future workplace needs • To understand the emerging trends aligning soft skills and training with industry needs • To emphasize the importance of adaptability and flexibility in the Future Workplace	Module V: TVET Future for Work (4 hours) 5.1. Concept of the Future of Work 5.2. Soft Skills for the Future Workplace 5.3. Emerging trends in TVET: <i>Green skills, digitalization, Entrepreneurial Mindset and Industry 4.0</i> 5.4. Adaptability and Flexibility in the Future Workplace

Major Teaching and Learning Strategy (List of some Teaching and Learning Strategies for the faculty)

- **Hybrid-based teaching:** Present information to students through oral presentation and deliver instructional content outside of class through videos or PowerPoint.
- **Inquiry-based and active learning:** Encourage students to ask questions and engage them to participate actively in the learning process. (For example: Group Discussion or Case Studies)

- **Problem-based, Project-based, and Collaborative learning:** Provide students with a project and coach them to apply knowledge and skills to understand, explore, and solve the problem collaboratively.
- **Reflective Practice:** Encourage students to reflect on their learning at the end of the module. Review of key learnings from the module along with the feedback on the module.

Practical (List some of the practical and field-based tasks to be done)

1. Field Visits: Students will have the opportunity to visit a range of institutions including TVET institutions, governmental agencies, and industry partners. During these visits, students will gain firsthand experience observing the implementation of TVET programs and policies. They will have the chance to witness the practical applications of what they have learned in the classroom and see how these programs and policies are being put into action. After each visit, students will be required to prepare a detailed field report that documents their observations, insights, and any recommendations they may have for improving the implementation of TVET programs and policies.

2. Case Studies: This project involves an in-depth exploration and analysis of the TVET ecosystems of specific countries. Students will be tasked with selecting a country and thoroughly examining its TVET ecosystem, including its policies, programs, and institutions. They will conduct research, analyze data, and present their findings in a comprehensive case study format. Through this project, students will gain a deeper understanding of the TVET landscape in different countries and develop critical thinking and research skills.

Assessment Plan

- **In-semester (50)**
Assignment, written exam, presentation, case study, report, objectives, attendance
- **End-semester (50)**

Question Types	Marks Distribution	Total Question
MCQ	10 Marks	10 Questions
SAQ	24 Marks	7 Questions [Attempt 6 Questions]
LAQ	16 Marks	3 Questions [Attempt 2 Questions]

Marking Scheme for Final Examination:

- Section A (**MCQ**): 10 Questions × 1 marks
- Section B (**Short Answer Question-SAQ**) – Total 7 Questions × 4 marks (Attempt any Six)
- Section C (**Long Answer Question-LAQ**) – Total 3 Questions × 8 marks (Attempt any Two)

Learning Units	Total Credit Hours	MCQs	SQs	LQs	Remarks
Module I: Introduction to TVET Education	4	1	1	0.5	
Module II: TVET Ecosystems Worldwide	6	2	2	0.5	

Module III: Framework of TVET in Nepal	10	3	2	1	
Module IV: TVET Human Resource Management	6	2	1	0.5	
Module V: TVET Future for Work	4	2	1	0.5	

References:

- Technical and Vocational Education and Training: Issues, Concerns and Prospects by UNESCO
- Skills for a Changing World: National Perspectives and the Global Movement by OECD
- Transforming Education: Learning Pathways in TVET by UNESCO-UNEVOC
- Technical and Vocational Education and Training in Nepal: Current Status and Future Directions by Government of Nepal, Ministry of Education.
- TVET Sector Strategic Plan (TSS) 2023 – 2032 AD
- TVET Reform Strategic Plan (TRSP) 2080 – 2089 BS

Occupational/Specialization Course

ITEX 300 Management Information System

Course Title: Management Information System

Course Code: ITEX 300

Credit Hours: 2 (1 TH +1 PR)

Nature of the course: Theory and Practical

Course Description and Goals

This course covers different concepts of Management Information System (MIS) applicable for Bachelor of Technical Education program which includes information system in the global industry, collaboration and tools, different types of information systems, ethical and social issues, concept of business intelligence, enterprise applications, and decision making. The objective of the course is to assist students integrate their learning from functional areas, decision making process in an organization and role of information. This course will prepare the students with understanding of role, advantages and components of an Information System.

Learning Outcomes

On completion of this course, the students will be able to:

- Develop knowledge of how Information Systems are used in organizations for meeting strategic and operational goals
- Acquire skills using current end-user software for communication, data transformation, collaboration, and problems solving
- Get knowledge on software and hardware components, information structures, basic business organization and processes, information system security, and networks
- Acquire concepts of management information system to manage business organizations and efficient decision making
- Align information systems with the objectives and processes of the organization
- Ability to function as an individual, and as a member or leader in diverse teams and in multi-disciplinary environment

Content with specific objectives

Specific objectives	Contents
• Analyze how information systems can enhance business processes, increase efficiency, and drive innovation • Explore the latest technologies and trends in information systems • Understand the ethical and legal issues related to information systems as well as data security and privacy.	Unit 1: Introduction of MIS (5 Hours) • Concepts of Information System • Evolution of MIS in organization • Role of data and information • Organization structures, Business Process, Systems Approach • Types of MIS and components of Information System (IS) • System Approach in Planning Organizing and Controlling MIS • Integration and automation of business functions and developing business models. Role and advantages of Transaction • Education Perspective on Information Systems
• Enhance efficiency and effectiveness of the	Unit 2: Information Systems, Organization, and Strategy (6 Hours)

organization. Improve effectiveness by providing employees with the information they need to make better decisions • Improve communication and collaboration in the organization • Facilitate the development, implementation, and monitoring of IT strategies that support the overall business strategy	• Impact of Information System on Organizations • Managerial Decision Making • Characteristics and components of Decision Support System. • Organization and Behavioral Impacts, Design and Understanding of Information Systems • Competitive Forces using Information Systems • Competitive Advantage and Business Value Chain Model • Business intelligence tools, and performance metrics • Decision Support Systems (DSS), Models and approaches to DSS
• Understand the role of IT infrastructure in supporting organizations and their operations. • Explore emerging technologies and their potential impact on businesses • Gain insights into the current trends and future developments in IT infrastructure and emerging technologies.	Unit 3: IT Infrastructure and Emerging Technologies (4 Hours) • IT infrastructure and evolution • Components of IT infrastructure • Trends in computer hardware platforms • Challenges of managing IT infrastructure and management solutions • Hardware, software, and networks • Telecommunications technologies used in MIS • Integrating business change with IT • Global IT infrastructure, and cross-cultural challenges in MIS implementation
• Understand the principles of effective information systems architecture and design in order to create systems that meet the needs of users and stakeholders • To stay current with emerging technologies and trends in the field of information systems architecture and design in order to remain competitive in the job market.	Unit 4: Architecture & Design of IS (5 Hours) • Architecture, development and maintenance of Information Systems, • Centralized and Decentralized Information Systems • Value and risk of IS • System Design: System design consideration, input/output design, forms design, file organization and database, data management, file design, program design and control
• Describe the importance of online marketing and advertising in the digital era.	Unit 5: E-Business and E-Commerce (5 Hours) • Features of e-commerce, digital markets, and digital goods

• Explain and exemplify the fundamental principles and concepts of e-business and e-commerce. • Learn how to effectively leverage technology to streamline business processes and operations	• Consumer and organization applications of e-Business • Principal e-commerce business and revenue models • Marketing with e-commerce • m-commerce in business, and the most important m-commerce applications • E-Commerce architecture
• Identify and formulate about the moral basis of individual responsibilities and actions, as well as the moral underpinnings of public policy. • Provide a space for students to develop awareness of different perspectives, reflect on their own views and biases and to think critically, question and take action on equality and justice issues in their communities	Unit 6: Ethical, Social and Security Issues in Information System (5 Hours) • Ethical and Social Issues of Information Systems Model and Moral Dimensions • Key Technology Trends on Ethical Issues, • Ethical Principles and Professional Codes of Conduct • Ethical Dilemmas • Information Right, Property Right, Accountability, Liability and System Quality • Security threats, vulnerabilities, controls, and risk management strategies.

Major Teaching and Learning Strategy (List of some Teaching and Learning Strategies for the faculty)

- Use real-world examples to illustrate how MIS is applied in different organizations.
- Assign students projects that require to use MIS software and tools to solve business problems.
- Guest speakers: Invite industry experts to speak to students about their experiences
- Group work and discussions
- Encourage students to actively engage by asking questions, participating in discussions, and completing hands-on activities.
- Enhance a collaborative learning environment to work together, share ideas
- Problem-based learning and set of technology

Practical (List some of the practical and field-based tasks to be done)

- Creating and maintaining databases
- Conducting data analysis
- Troubleshooting technical issues
- Developing and implementing new systems
- Training and supporting users
- Managing security
- Conducting research

Assessment Plan

- **In-semester (50)**
Assignment, written exam, presentation, case study, report, objectives, attendance
- **End-semester (50)**

Question Types	Marks Distribution	Total Question
MCQ	10 Marks	10 Questions
SAQ	24 Marks	7 Questions [Attempt 6 Questions]
LAQ	16 Marks	3 Questions [Attempt 2 Questions]

Marking Scheme for Final Examination:

- Section A (**MCQ**): 10 Questions × 1 marks
- Section B (**Short Answer Question-SAQ**) – Total 7 Questions × 4 marks (Attempt any Six)
- Section C (**Long Answer Question-LAQ**) – Total 3 Questions × 8 marks (Attempt any Two)

Learning Units	Total Credit Hours	MCQs	SQs	LQs	Remarks
Introduction of MIS	5 Hours	1	1	-	
Information Systems, Organization, and Strategy	6 Hours	3	2	1	
IT Infrastructure and Emerging Technologies	4 Hours	2	1	1	
Architecture & Design of IS	5 Hours	1	1	-	
E-Business and E-Commerce	5 Hours	2	1	1	
Ethical, Social and Security Issues in Information System	5 Hours	1	1	-	

References:

1. O'Brien, J., & Marakas, G. (2010). *Management information system* (10th ed.). McGraw Hill.
2. Laudon, K. C., & Laudon, J. P. (2022). *Management information systems: Managing the digital firm* (17th ed.). Pearson.

PDF Books

1. Edusat Learning Resource Material on Management Information System. Available online at: <chrome-extension://efaidnbmnnnibpcajpcglclefindmkaj/https://www.sctevtservices.nic.in/docs/website/pdf/140339.pdf>
2. Management Information Systems. Lovely Professional University. Available online at: chrome-extension://efaidnbmnnnibpcajpcglclefindmkaj/https://ebooks.lpude.in/management/mba/term_4/DMGT505_MANAGEMENT_INFORMATION_SYSTEM.pdf
3. Management Information Systems. Kenneth C. Laudon (New York University), Jane P. Laudon. Pearson Education Limited. Available online at: chrome-extension://efaidnbmnnnibpcajpcglclefindmkaj/https://repository.dinus.ac.id/docs/ajar/Kenneth_C.Laudon,Jane_P_.Laudon_-_Management_Information_Sysrem_13th_Edition_.pdf

ITEX 301 System Network Administration

Course: System Network Administration

Course Code: ITEX 301

Credit Hours: 2 (1 TH + 1 PR)

Nature: Theory and Practical

Course Description and Goals

The System and Network Administration course comprehensively introduces managing computer systems and networks. Students will explore the fundamental concepts, challenges, and best practices in system administration, including Unix-like and Windows systems, Linux distributions, and shell commands. The course covers account and security administration, file system management, logical volume management, and RAID configurations. Network management topics include TCP/IP networking, configuring various network services like web, DNS, mail, and proxy servers, and troubleshooting network issues. Students will also learn the installation and management of application servers, system security measures, and analytical techniques for system observation and problem-solving. The course integrates theoretical knowledge with practical lab sessions to develop hands-on skills in system and network administration, preparing students for roles in IT infrastructure management and support.

Learning Outcomes

On completion of this course, the students will be able to:

- Gain a deep understanding of system administration principles, including user and group management, file system administration, and disk management, enabling them to effectively configure and maintain computer systems.
- Through hands-on experience and theoretical knowledge, students will develop expertise in network administration, including TCP/IP networking, configuring network services like web servers, DNS servers, and mail transfer agents, and troubleshooting network issues.
- Acquire skills in managing system and network security, including implementing access controls, setting up firewalls, and configuring security measures such as TCP wrappers and iptables, ensuring the integrity and confidentiality of IT systems.

Content with specific objectives

Specific objectives	Contents
• Explain and describe the objectives, challenges, and common practices in system and network administration. • Differentiate between UNIX-like and Windows systems, including their core functionalities and user interfaces. • Master essential and advanced shell commands for efficient system navigation and administration.	Unit 1: Introduction to System & Network Administration (3 hours) • Objectives/Goals, Challenges, and Common Practices • Overview of the Operating Systems • Unix-like Systems Vs. Windows Systems • Linux Distributions and User Interfaces • Linux Operations Review • File System Hierarchy and Standard • Single-rooted hierarchy, Seamless and Extensible File systems • Mounting Additional File systems • File system Object-Oriented Design and File system Standard • Unix File and Directory Permissions • Essential Shell Commands • Basic File Manipulation Commands and Directory Navigation Commands

	- Advanced File Manipulation Commands (Init, Processes, and Threads) - Advanced Shell Features
- Demonstrate proficiency in user and group management, including user-private group schemes. - Apply knowledge of password aging, default user files, and disk quotas in managing user accounts. - Implement effective file and folder permissions, ownership management, and access controls.	Unit 2: Account and Security Administration (3 hours) - User and Group Concepts, and User Private Group Scheme - User Administration, Modifying Accounts, and Group Administration - Password Aging and Default User Files - Managing Files and Folder Permissions - Managing File Ownership - Controlling Access to Files - Managing Disk Quotas
- Demonstrate proficiency in user and group management, including user private group schemes. - Apply knowledge of password aging, default user files, and disk quotas in managing user accounts. - Implement effective file and folder permissions, ownership management, and access controls.	Unit 3: File System Administration (4 hours) - Partitioning Disks with fdisk and parted - Creating, Mounting, and Maintaining File systems - Swap - Determining Disk Usage With df and du - Configuring Disk Quotas - Logical Volume Management (LVM) and RAID - Implementing LVM, Creating Logical Volumes (LVs), Manipulating VGs & LVs - Advanced LVM Concepts (system- config-lvm) - RAID Concepts (Creating and Managing a RAID-5 Array)
- Configure TCP/IP networking, DNS servers, web servers, and mail transfer agents for efficient network services. - Troubleshoot network issues using tools like ping, traceroute, netstat, and analyze network traffic using proxy caches. - Implement secure remote administration using SSH, SCP, and configure firewall rules for network security.	Unit 4: Network Management (6 hours) - TCP/IP Networking - Configuring a Linux Box for Networking - Configuring a Linux Box as a Router - Configuring a Web Server (Apache) - Configuring a DNS Server (BIND) - Configuring Mail Transfer Agents (Postfix) - Configuring Proxy Caches (Squid) - Network Configuration (IP Networking and Linux Network Configuration) - Network Services - Dynamic Host Control Protocol (DHCP) - Network Time Services and Sharing Desktops with VNC - RPC-Based Services and INET Super Server - TCP/IP Troubleshooting: ping, traceroute, ifconfig, netstat, ipconfig - Remote Administration with SSH and SCP - Configuration, Telnet Replacement, Secure Copy and Rsync

	• RSA and DSA Authentication (Password-less Logins) • Remote Command Execution and Port Forwarding
• Install and configure network services like DHCP, DNS, FTP, and NFS for efficient server management. • Set up secure communication using Open SSH and manage file sharing between Linux and Windows systems using SAMBA. • Implement security measures such as firewalls and web servers to protect network resources.	Unit 5: Installation of Application Server and Management (6 hours) • DHCP, DNS, Telnet server; compare with other NOS setups of corresponding network services • Open SSH: Secure Network Communication • FTP and setting up Mail Servers and Client • Network Information Service (NIS) and Sharing File systems (NFS) • Linux and Windows File and Printer Sharing • Tracking Domain Names and Address • Setting up a Firewall and a Web server
• Develop strategies for troubleshooting common system and network problems, including boot issues and data restoration. • Utilize event viewer tools to monitor system performance and connectivity, ensuring smooth network operations. • Implement backup and restore procedures for system and data recovery in case of failures.	Unit 6: Managing Network Services (4 hours) • Maintenance Troubleshooting: Common System and Network Problems • Developing General Strategies • Resolve Boot Problems, Backup and Restore Data and System Volume • Using Event Viewer and Troubleshoot Connectivity
• Implement security measures such as login security, TCP wrappers, and iptables to protect against unauthorized access. • Configure firewall rules for packet filtering, port forwarding, and IP masquerading to secure network communications. • Detect and respond to intrusions using intrusion detection systems and implement mandatory access controls for system security.	Unit 7: Systems Security (4 hours) • Overview, Application Security, and Login Security • Boot Loader Security (LILO and GRUB) • TCP Wrappers Configuration • Iptables Firewalling: Preliminaries • Iptables Scenarios • Packet Filtering • Port-Forwarding/Redirection and NAT/IP Masquerading • Packet-Processing Model • Intrusion Detection and Mandatory Access Control (MAC)

Major Teaching and Learning Strategy (List of some Teaching and Learning Strategies for the faculty)

- Lecture Sessions: Traditional lectures delivered by instructors to introduce concepts, theories, and best practices in system and network administration. These sessions can provide a foundational understanding of the subject matter.
- Hands-on Labs: Practical laboratory sessions where students engage in real-world tasks such as system setup, configuration, troubleshooting, and security implementation.

Labs allow students to apply theoretical knowledge in a practical setting, enhancing their technical skills.

- **Case Studies:** Analyzing real-world case studies related to system and network administration challenges. Case studies provide a context for understanding complex issues, decision-making processes, and effective problem-solving strategies.
- **Group Projects:** Collaborative projects where students work in teams to design, implement, and present solutions to system and network administration problems. Group projects promote teamwork, communication skills, and practical application of course concepts.
- **Simulation Exercises:** Using simulation software or virtual environments to simulate network configurations, security scenarios, and system performance evaluations. Simulations offer a risk-free environment for hands-on practice and experimentation.
- **Guest Lectures and Industry Talks:** Inviting industry experts or guest speakers to share their experiences, insights, and best practices in system and network administration. Guest lectures provide real-world perspectives and networking opportunities for students.
- **Online Resources:** Utilizing online platforms, learning management systems, and digital resources for self-paced learning, supplemental materials, interactive tutorials, and practice exercises. Online resources enhance accessibility and flexibility in learning.
- **Assessment Methods:** Employing a variety of assessment methods such as quizzes, exams, lab reports, presentations, and projects to evaluate students' understanding, practical skills, critical thinking, and problem-solving abilities.

Practical (List some of the practical and field-based tasks to be done)

- Overview of Operating Systems Installation and Setup
- File System Navigation and Basic Shell Commands Practice
- User and Group Management in Linux
- File Permissions and Access Control Testing
- Disk Partitioning and File System Creation
- Logical Volume Management (LVM) Setup and RAID Configuration
- TCP/IP Network Configuration and Troubleshooting
- Web Server (Apache) and DNS Server Configuration
- DHCP, DNS, and Mail Server Setup
- Firewall Configuration and Network Security Testing
- Troubleshooting Network Issues and Connectivity Problems
- Backup and Restore Procedures Implementation
- Security Measures Setup (Firewalls, VPNs, Access Controls)
- Intrusion Detection and Incident Response Simulation

Assessment Plan

- **In-semester (50)**
Assignment, written exam, presentation, case study, report, objectives, attendance
- **End-semester (50)**

Question Types	Marks Distribution	Total Question
MCQ	10 Marks	10 Questions
SAQ	24 Marks	7 Questions [Attempt 6 Questions]
LAQ	16 Marks	3 Questions [Attempt 2 Questions]

Marking Scheme for Final Examination:

- Section A (**MCQ**): 10 Questions × 1 marks
- Section B (**Short Answer Question-SAQ**) – Total 7 Questions × 4 marks (Attempt any Six)
- Section C (**Long Answer Question-LAQ**) – Total 3 Questions × 8 marks (Attempt any Two)

Learning Units	Total Credit Hours	MCQs	SQs	LQs	Remarks
Introduction to System & Network Administration	3	1	1		
Account and Security Administration	3	1	1		
File System Administration	4	2	1	1	
Network Management	6	2	1	1	
Installation of Application Server and Management	6	2	1	1	
Managing Network Services	4	1	1		
Systems Security	4	1	1		

References:

- Limoncelli, T. A., Hogan, C. J., & Chalup, S. R. (2016). *The practice of system and network administration*. Addison-Wesley Professional.
- Nemeth, E., Snyder, G., Hein, T. R., & Whaley, B. (2010). *UNIX and Linux system administration handbook* (4th ed.). Prentice Hall.
- Nemeth, E., Snyder, G., Hein, T. R., Whaley, B., & Mackin's, D. (2017). *UNIX and Linux system administration handbook* (4th ed.). Addison-Wesley Professional.
- Donahue, G. A. (2011). *Network warrior* (2nd ed.). O'Reilly Media.

ITEX 320 Web Engineering

Course Title: Web Engineering

Course Code: ITEX 213320

Credit Hours: 3 (2 TH + 1 PR)

Nature: Theory +Practical

Course Description and Goals

This course provides an in-depth exploration of systematic web application development. Students will learn advanced topics in web engineering, such as requirements engineering, web application modeling, architecture, security, performance optimization, and maintenance. The course will include a mix of theoretical concepts and practical projects to develop robust, scalable, and secure web applications. The prerequisite for this course is the course ITEX 213 Principles of Internet Technologies and Web Applications.

Learning Outcomes

On completion of this course, the students will be able to:

1. Apply systematic engineering principles to design and develop web applications.
2. Analyze and model web applications using advanced tools and methodologies.
3. Evaluate and optimize the architecture of web applications for performance and scalability.
4. Implement comprehensive security measures in web applications.
5. Manage the operation and maintenance of web applications effectively.

Content with specific objectives

Specific objectives	Contents
• Identify and Describe Key Components of Web-based Systems. • Describe/interpret and Analyze the Principles of Web Engineering • Describe/interpret Agile Methodologies to Web Development Scenarios	Unit 1: Introduction to Web Engineering (2 hours) • Web-based system: The Web, WebApps, WebApp Attributes, WebApp Types • Web Engineering: Introduction, Agile, Agile Process and Principles, WebE Framework, WebE Activities, WebE Methods, Best Practices.
• Analyze the Incremental Process Flow and WebE Process Activities. • Evaluate the Role and Impact of Umbrella Activities in WebE	Unit 2: Web Engineering Process (2 hours) • Incremental process flow, WebE Process Activities and actions • Umbrella Activities
• Demonstrate the ability to conduct elicitation tasks to gather information effectively from various sources. • Analyze user descriptions to accurately determine content needs and desired functionality. • Describe/interpret the fundamental concepts and techniques of negotiation.	Unit 3: Communication phase (4 hours) • Communication Activities, Formulation • Elicitation: Elicitation tasks, User descriptions, content and functionality, constraints and performance, capturing interaction: use cases, negotiation

• Describe/interpret the fundamental principles and purposes of various modeling frameworks. • Describe/explain and explain different methodologies used in web development, including Agile	Unit 4: Modeling phase (4 hours) • Introduction, Modeling Frameworks, Modeling Languages • Web application modeling, web methodologies
• Analyze user needs to inform the design and development of user-centric systems. • Evaluate web content models for their efficiency in managing and conveying information within a web environment. • Apply the concept of a data tree to organize and represent information structure on web platforms. • Develop sequence diagrams, activity diagram, state diagram to model the flow of interactions in a system accurately.	Unit 5: Analysis Modeling (4 hours) • Introduction, Analysis outputs, Understanding users, • Content Model: Web information exchange, Data tree • Interaction Model: Sequence diagram, state diagram, • Functional Model: Activity diagram • Configuration Model
• Identify and articulate clear design goals for a web application project based on stakeholder requirements and user research. • Analyze the relationship between design choices and web application quality, focusing on performance, scalability, and user satisfaction. • Synthesize requirements and design elements to refine the conceptual, technical and developing architecture of a web application.	Unit 6: WebApp Design (4 hours) • Design for WebApp • Design Goals • Design and WebApp Quality • Design Actions • Design Process: Conceptual Architecture, Developing architecture, Technical Architecture
• Describe/explain and describe the foundational principles that guide effective design across various disciplines. • Apply usability principles to design user-centric products and interfaces.	Unit 7: Interface Design (4 hours) • Introduction, design principles, pragmatic design guidelines, workflow, design steps, Aesthetic design, usability, design issues.

- Describe/explain the key characteristics of information architecture and its role in organizing and structuring information. - Design wireframe models that provide a visual guide for the layout and interaction patterns of a web application or site. - Compare the Relationship Management Methodology (RMM) models with other design approaches for navigating structures.	Unit 8: Information Design (6 hours) - Information Architecture and its characteristics, good structure, blueprints, assessing information, understanding context, context guidelines, - Wireframe models - Navigating design: RMM models and other approaches
- Describe/explain the role of architecture in web design, including how it supports scalability, maintainability, and user experience. - Explain the Model-View-Controller (MVC) design pattern and its components.	Unit 9: Functional Design (2 hours) Categories and examples, Architecture, MVC design pattern, WebML, State modeling.
- Describe/explain the significance and overarching process of construction and deployment in the web development lifecycle. - Demonstrate the ability to select appropriate tools, frameworks, and methodologies for the construction of software based on project requirements. - Demonstrate the steps of deploying a web application in a consistent and reliable manner.	Unit 10: Construction and Deployment (6 hours) - Introduction - Construction principles: preparation principles, selection principles, coding principles, CM principles - Deployment and its principles - Use of components, design steps
- Describe/explain and implement a systematic testing process that includes planning, execution, and documentation. - Describe/explain and perform different testing to ensure the accuracy, relevance, and timeliness of web application content.	Unit 11: Tools and Technologies, and Testing WebApps (7 hours) - Introduction, application framework, content management, Development tools, - Dimensions of Quality - Testing strategy, Testing process, content testing, User interface Testing, Usability Testing, Compatibility Testing, Navigation

	Testing, Configuration Testing, Security Testing, Performance testing
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Major Teaching and Learning Strategy (List of some Teaching and Learning Strategies for the faculty)

- Lecture
- Project-based learning to encourage students to engage in hands-on projects that require applying the principles and tools learned to create real or simulated web applications.
- Guest Lecture from Industry: Incorporating guest lectures from industry professionals into a course can greatly enrich the learning experience for students.
- Flipped Classroom (if possible) to have students prepare by watching lectures or reading material at home, then use class time for discussions, exercises, and applying concepts in a workshop setting.

Practical (List some of the practical and field-based tasks to be done)

Assessment Plan

- **In-semester (50)**
 1. Assignments
 2. Written exam (First Internal Exam, Second Internal Exam)
 3. Presentations on different subtopics under each unit (Group work)
 4. Case study (group work)
 5. Mini project (group work)
 - a. Report/Documentation
 - b. Presentation

- **End-semester (50)**

Question Types	Marks Distribution	Total Question
MCQ	10 Marks	10 Questions
SAQ	24 Marks	7 Questions [Attempt 6 Questions]
LAQ	16 Marks	3 Questions [Attempt 2 Questions]

Marking Scheme for Final Examination:

- Section A (**MCQ**): 10 Questions × 1 marks
- Section B (**Short Answer Question-SAQ**) – Total 7 Questions × 4 marks (Attempt any Six)
- Section C (**Long Answer Question-LAQ**) – Total 3 Questions × 8 marks (Attempt any Two)

Learning Units	Total Credit Hours	MCQs	SAQ	LAQ	Remarks
Unit 1: Introduction to Web Engineering	2	1	1	0	

Unit 2: Web Engineering Process	2	1	1	0	“1” represents that Question can be asked in that section. “0” represents that it is better not to include that Unit in the specific section.
Unit 3: Communication Phase	4	1	1	1	
Unit 4: Modeling Phase	4	1	1	1	
Unit 5: Analysis Modeling	4	1	1	1	
Unit 6: WebApp Design	4	1	1	1	
Unit 7: Interface Design	4	1	1	1	
Unit 8: Information Design	6	1	1	1	
Unit 9: Functional Design	2	1	1	1	
Unit 10: Construction and Deployment	6	1	1	1	
Unit 11: Tools and Technologies, and Testing WebApp	7	1	1	1	

References:

Kappel, G., Proll, B., Reich, S., & Retschitzegger, W. (2006). *Web engineering: The discipline of systematic development of web applications*. John Wiley & Sons, Ltd.

ITEX 321 Software Engineering

Course Title: Software Engineering

Credit Hours: 3 (2 TH + 1 PR)

Code: ITEX 321

Nature: Theory +Practical

Course Description and Goals

This course deals with software and software development methodologies. This course focuses on the knowledge of software engineering and its application in the real life for software development. It is concerned with theories, methods, and tools which are needed to develop computer software.

Learning Outcomes

On completion of this course, the students will be able to:

- Identify real life applications of compute-based system.
- Develop professional software engineering and software engineering skills.
- Differentiate different software development process models
- Apply requirement engineering process to collect functional and non-functional requirements.
- Take decision to use different architectural designs.
- Use different testing methodologies while developing software.

Content with specific objectives

Specific objectives	Contents
○ Describe and exemplify what software engineering is and why it is important; ○ understand that the development of different types of software system may require different software engineering techniques; ○ Describe and exemplify ethical and professional issues that are important for software engineers; ○ have been introduced to four systems, of different types, which are used as examples throughout the book.	Unit 1: Introduction to Software Engineering [5 Hrs.] • Professional software development • Software engineering ethics • Case studies
○ Interpret the concepts of software processes and software process models; ○ have been introduced to three general software process models and when they might be used; ○ Describe and exemplify about the fundamental process activities of software requirements engineering, software development, testing, and evolution; ○ Describe and exemplify why processes should be organized to cope with changes in the software requirements and design; ○ Describe and exemplify the notion of software process improvement and the factors that affect software process quality.	Unit 2: Software Processes [5 Hrs.] • Software process models • Process activities • Coping with change • Process improvement

○ Describe and exemplify the rationale for agile software development methods, the agile manifesto, and the differences between agile and plan-driven development; ○ know about important agile development practices such as user stories, refactoring, pair programming and test-first development; ○ Interpret and exemplify the Scrum approach to agile project management; ○ Interpret and exemplify the issues of scaling agile development methods and combining agile approaches with plan-driven approaches in the development of large software systems.	Unit 3: Agile Software Development [8 Hrs.] ● Agile methods ● Agile development techniques ● Agile project management ● Scaling agile methods
○ Describe and exemplify the concepts of user and system requirements and why these requirements should be written in different ways; ○ Describe and exemplify the differences between functional and non-functional software requirements; ○ Describe and exemplify the main requirements engineering activities of elicitation, analysis, and validation, and the relationships between these activities; ○ Describe why requirements management is necessary and how it supports other requirements engineering activities.	Unit 4: Requirements Engineering [10 Hrs.] ● Functional and non-functional requirements ● Requirements engineering processes ● Requirements elicitation ● Requirements specification ● Requirements validation ● Requirements change
○ Describe and exemplify how graphical models can be used to represent software systems and why several types of model are needed to fully represent a system; ○ Interpret the fundamental system modeling perspectives of context, interaction, structure, and behavior; ○ Interpret the principal diagram types in the Unified Modeling Language (UML) and how these diagrams may be used in system modeling; ○ have been introduced to model-driven engineering, where an executable system is automatically generated from structural and behavioral models.	Unit 5: System Modeling [10 Hrs.] ● Context models ● Interaction models ● Structural models ● Behavioral models ● Model-driven engineering
○ Describe the stages of testing from testing during development to acceptance testing by system customers; ○ have been introduced to techniques that help you choose test cases that are geared to discovering program defects; ○ Evaluate and Describe test-first development, where you design tests before writing code and run these tests automatically;	Unit 6: Software Testing [7 Hrs.] ● Development testing ● Test-driven development ● Release testing ● User testing

○ Interpret about three distinct types of testing-component testing, system testing, and release testing; ○ Differentiate between development testing and user testing.	
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Major Teaching and Learning Strategy

- ✓ Lecture classes
- ✓ Demonstration
- ✓ Regular assessments
- ✓ Presentation
- ✓ Case study

Project and Practical

Students need to form a group and conduct a mini-project. This includes identifying a real problem and implementing several concepts discussed in the classroom to develop simple software. This is a group work. Hence, a collaborative approach will be encouraged. Some examples include developing Online Shopping Software, Library Management Systems, Inventory Management Systems, Student Profile Systems, Seat Reservation Systems, eSewa, Khalti, etc.

Assessment Plan

- In-semester (50)
Assignment, written exam, presentation, project/case study/report, objectives, class performance/attendance
- End-semester (50)

Question Types	Marks Distribution	Total Question
MCQ	10 Marks	10 Questions
SAQ	24 Marks	7 Questions [Attempt 6 Questions]
LAQ	16 Marks	3 Questions [Attempt 2 Questions]

Marking Scheme for Final Examination:

- Section A (MCQ): 10 Questions × 1 marks
- Section B (Short Answer Question-SAQ) – Total 7 Questions × 4 marks (Attempt any Six)
- Section C (Long Answer Question-LAQ) – Total 3 Questions × 8 marks (Attempt any Two)

Learning Units	Total Credit Hours	MCQs	SQs	LQs	Remarks
Unit 1: Introduction to Software Engineering	5	1	1		
Unit 2: Software Processes	5	1	1		
Unit 3: Agile Software Development	8	2	1	1	
Unit 4: Requirements Engineering	10	2	1	1	
Unit 5: System Modeling	10	2	1	1	
Unit 6: Software Testing	7	2	2		
Total	45	10	7	3	

Reference Materials:

- Sommerville, I. (2015). *Software engineering* (10th ed.). Pearson.
- Pressman, R., Maxim, B. R. (2019). *Software engineering: A practitioner's approach* (9th ed.). MacGraw-Hill Education.

ITEX 340 Professional IT Project I

Course Title: Professional IT Project I

Course Code: ITEX 340

Credit Hours: 2 (Practical)

Nature: Practical

Course Description and Goals

This project-based course emphasizes collaborative problem-solving in Information Technology (IT) within local educational challenges. Each student group will be assigned a supervisor to guide their progress, while an external evaluator will assess the final project outcomes through inquiry, viva, and demonstration. Additionally, students will maintain individual log sheets to track their contributions, progress, and reflections throughout the project. **Theme:** Project-based learning with Supervised Progress

Learning Outcomes

On completion of this course, the students will be able to:

- Identify and analyze local educational challenges suitable for IT-based solutions.
- Formulate project proposals with clear objectives, methodologies, and expected outcomes.
- Apply IT concepts and tools to design, develop, and implement innovative solutions.
- Collaborate effectively in groups, demonstrating teamwork, communication, and project management skills.
- Present project findings professionally and respond to inquiries and feedback during viva and demonstration.
- Reflect on the project experience, lessons learned, and implications for future IT projects.

Course Expectations:

1. Group Formation and Project Proposal:
 - a. Form groups based on common interests and skills, and submit a detailed project proposal outlining objectives, methodologies, and timelines.
 - b. Each group will be assigned a supervisor for ongoing guidance and progress monitoring.
2. Weekly Progress Updates with Supervisor:
 - a. Groups are required to contact their supervisors weekly to discuss progress, and challenges, and receive feedback.
 - b. Supervisors will provide input and guidance to ensure projects stay on track and meet objectives.
3. Individual Log Sheet Maintenance:
 - a. Students will maintain individual log sheets to document their contributions, tasks completed, challenges faced, and reflections on the project.
 - b. Log sheets should be updated regularly and submitted as part of the final project documentation.
4. Project Development and Implementation:
 - a. Utilize IT tools and techniques to design, develop, and implement solutions for identified educational challenges.
 - b. Regularly communicate with supervisors to showcase progress and seek input.

5. Project Presentation and Demonstration:
 - a. Prepare a comprehensive project presentation and demonstration showcasing key features, functionality, and outcomes.
 - b. Respond to inquiries and feedback during viva and demonstration sessions conducted by the external evaluator.
6. Project Report and Reflection:
 - a. Submit a detailed project report documenting the problem-solving process, methodologies, results, and recommendations.
 - b. Reflect on the project experience, challenges faced, lessons learned, and future implications for IT projects in the individual log sheets.

Course Structure:

- Group Formation and Project Proposal Submission - Weeks 1-2
- Weekly Progress Updates with Supervisors - Weeks 3-13
- Project Development and Implementation - Weeks 4-14
- Project Report Submission and Reflection - Week 15
- Project Presentation and Demonstration - Week 15
- Individual Log Sheet Maintenance - Throughout the Course

Assessment Methods:

Supervisor Evaluation (Fortnightly Progress) - 60%

External Evaluation (Viva, Demonstration, Project Report) - 40%

Note: The course syllabus and assessment weights are subject to adjustments based on project progress, feedback, and instructor discretion. Individual log sheets will contribute to the overall assessment through reflection and documentation of individual contributions to the project. An external examiner will be from Kathmandu University.

Table 1: Course Structure Gantt chart

	Project Proposal and Group Formation		Project Development and Implementation											Project Presentation and Documentation		External Feedback and Grading
Tasks /Weeks	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Group Formation	✓	✓														
Project Proposal	✓	✓														
Supervisor Meetings			✓		✓		✓		✓		✓		✓		✓	
Individual Log			✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
Project Development			✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
Presentation Preparation													✓	✓	✓	
Project Report													✓	✓	✓	
Internal Reflection														✓	✓	
External Feedback & Grading																✓